

Immanuel Annoh

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PROFESSIONAL SUMMARY

Data Analyst / Data Scientist with MS in Data Science specializing in predictive modeling, causal inference, and BI dashboards across healthcare and retail analytics. Built automated SQL/Python reporting pipelines and deployed Fast API inference services to support operational decisions.

TECHNICAL SKILLS

- Programming & Analysis: Python, R, SQL (joins, window functions, subqueries)
- Visualization & BI: Tableau, Power BI, Excel (advanced formulas, dashboards), ggplot2, Folium
- Databases & ETL: PostgreSQL, MySQL, SQLite; Power Query; data ingestion, transformation
- Methods: Regression, classification, clustering, hypothesis testing/A-B testing, time series, feature engineering, KPI design
- Deployment & Tools: Fast API, Docker, Git/GitHub, RStudio

PROFESSIONAL EXPERIENCE

Data Analyst | Decabyte (Tech Services) | Remote **09/2023 – Oct 2025**

- Inventory Optimization: Engineered a Python and SQL-based inventory analytics system, resulting in a 25% reduction in stockouts and a 30% increase in customer satisfaction by aligning supply with demand forecasts.
- Pipeline Automation: Automated end-to-end ETL reporting pipelines (daily/weekly/monthly), saving 10+ hours of manual labor per week and eliminating human error in data processing.
- Dashboard Development: Designed executive-level Power BI dashboards to track KPIs, providing leadership with real-time visibility into sales performance and forecasting accuracy.

Data Entry Specialist | Decabyte | Remote **08/2019 – 09/2023**

- Data Quality Assurance: Managed high-volume operational datasets, performing rigorous quality checks and validation to ensure 99% data accuracy for downstream analytics.
- Process Improvement: Identified inconsistencies in legacy data systems and standardized entry protocols, improving data reliability for the operations team.
- Reporting Support: Prepared clean, structured datasets for internal reports, adhering to strict data confidentiality and security standards.

PROJECTS

Hospital Readmission Risk Modeling (End-to-End ML Pipeline) **07/2025**

- Objective: Reduce healthcare costs by predicting 30-day patient readmission risks using a dataset of 1M patient encounters.
- Solution: Built a leakage-safe XGBoost and Logistic Regression pipeline. Applied Difference-in-Differences for causal inference and SHAP values to explain predictions to clinical stakeholders.
- Deployment: Deployed the model as a production-ready Fast API service, enabling real-time risk scoring with calibrated probabilities.

Airbnb Booking Conversion Predictor **03/2025**

- Objective: Identify key behavioral drivers to improve booking completion rates.
- Solution: Developed an XGBoost conversion model on 50K booking records, successfully handling high-cardinality features and class imbalance.
- Impact: Utilized SHAP analysis to isolate key features, providing marketing teams with actionable data to target high-intent users.

EDUCATION

- Master of Science, Data Science | GPA: 3.9/4.0 | Eastern University | St. Davids, PA | 03/2025
- Bachelor of Science, Biology | Siena College | Loudonville, NY | 05/2022

CERTIFICATION

- IBM DATA SCIENCE PROFESSIONAL CERTIFICATE